CMSC 5718 Introduction to Computational Finance

Assignment 1: Derivative pricing and hedging (20% of total grade)

Instructions

1) Use Python/Jupyter to complete the assignment. Give the numerical answers to each part.
2) Submit a copy of your report together with supporting programs and/or data files (as a zipped file) by **uploading to Blackboard on or before February 23, 2026, 11:59pm.** The **file name** of the zipped file or your report should include your surname and have the following format, **e.g. ChauKL_Assign1**. [If uploading to Blackboard is not successful, you may consider sending a email to kalokchau@cuhk.edu.hk, but submission through Blackboard is preferred.]
3) Late submission of one week or less will attract a penalty of 20% of the assignment mark. ***Submission will not be accepted after March 2, 2026, 11:59pm.***
4) You can either submit your work **individually** or work together in a **group of two or three students**. Students in the same group will get the same assignment grade. Please state the name(s) and student number(s) clearly in the report. However, if you are submitting as a group, you only need to submit one copy of the report with one of the names in the file name (described in (1) above).
5) ***Please observe the university's plagiarism guidelines.***

Introduction

In the first part of this assignment, we will test the implementation of an option pricing model using the Monte Carlo method. It will be applied to the pricing of an equity structured product. The second part will test the effectiveness of the delta hedging scheme under the Black-Scholes framework.

Choose the stock that you have to work on

Use your student number to decide which stocks you have to use to perform the analysis. Take the last digit of your student number to obtain the order number, and look up the stock codes and relevant data from the given data sheet. For example, if your student number ends with 4, the order number is 4, and the stocks are CNOOC (S_1, stock code 883) and Sinopec (S_2, stock code 386). If you work in a group, select the order number based on one of your student numbers. In question 2, use the two stocks S_1 and S_2. In questions 1 and 3, the calculation should be based on S_1 only.

Note: Dividends, transaction costs and board lots can be ignored in this exercise.

1. European Option pricing (20%)

i) Use the Black-Scholes model to price the following option for stock S_1:

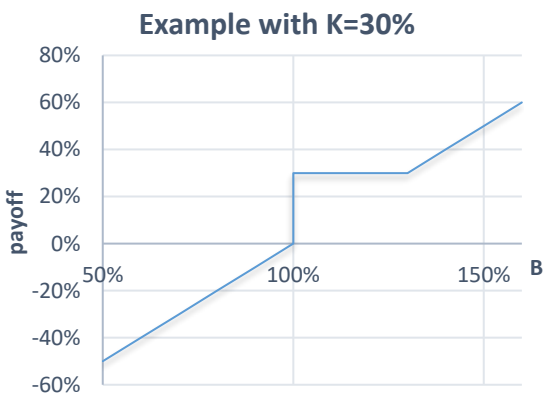
European call option, at-the-money (S=K=price of the stock as of 30 June, 2025), continuously compounded interest rate r = 3.14%p.a., maturity T = 0.5014 year.

ii) Use a Monte Carlo scheme with time steps N = 120, i.e. Δt = T/N = 0.004178 (refer to the discretization scheme in Topic 1-2, slides 37 and 38; don't use the exact formula). Give the answers with: (a) 10000 paths; (b) 500000 paths. Record the computation times for these runs.

[Hint: the answers for (ii)(a) and (ii)(b) should be close to the answer in (i) above]

2. Exotic Option Pricing (35%)

An *exotic option* with two stocks S_1 and S_2 is based on the following conditions (payable at maturity t = T = 0.5 year from start date):



$S_{1,0}$, $S_{2,0}$ = stock prices at time t=0;
$S_{1,1}$, $S_{2,1}$ = stock prices at time t=0.5 year;
B = min($S_{1,1}$ / $S_{1,0}$, $S_{2,1}$ / $S_{2,0}$);
Payoff = B–1, if B < 1 or B > 1+K;
= K, if 1 ≤ B ≤ 1+K
min(A,B) returns the minimum value of A and B.

Find K such that the fair price of the product as of the start date (time t=0) is –0.03. Use a Monte Carlo scheme with one time step (refer to the discretization scheme in Topic 1-2, slides 40, 41, 43; use 1000000 paths). Continuously compounded interest rate r = 3.14% p.a. Use the parameters given in the spreadsheet for the stock pair, and assume a start date of June 30, 2025.

3. Test of delta hedging strategy (45%)

i) Assume that you are **<u>short</u>** M call options on S_1 as described in Q1(i), as of June 30, 2025. Using the daily price data and the implied volatility given in the spreadsheet, construct a delta hedging strategy for the position for the period between June 30, 2025 to December 30, 2025, so that the overall position is delta neutral daily (the format is given in the spreadsheet). The account balance in each day is calculated by summing the following components:

- Previous account balance.
- The account balance is negative, and interest has to be paid. The interest is calculated daily, and the interest amount = previous day amount balance x [exp(interest_rate x num_days/365) $-$ 1], where num_days is the number of days from the previous date to the current date [it can be one day or more than one day, depending on whether there are holidays].
- Cash required / received from share transaction.

In your report, include a few lines from these tables (but no need to include all the dates). On maturity date, if the final share price is above the strike price, the call would be exercised. The number of shares in the share account must be equal to +M, and this position is to be sold to the option holder **at the strike price**. Denote F_{i} as the final account balance after taking into account of the above transactions. Find F_{i}. [This should be a negative amount]

ii) On June 30, 2025, you deposited the money that you received from shorting M call options into a deposit account, earning a continuously compounded interest of 3.14% p.a. The maturity of the deposit is December 30, 2025, and the total amount received is P_{i}. Find P_{i}. Compare P_{i} with F_{i} obtained in Q3(i). Does the final account balance F_{i} match the corresponding total amount in the deposit account P_{i} (noting that one is positive and the other is negative)? Comment briefly on the result.